

LLM-Enhanced Dynamic Graph Networks for CVR Prediction

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Supervisor Meeting Reference

1. What Problem Does This Dissertation Address?

- In e-commerce advertising, users click ads but may not purchase immediately — conversion can happen days later.
- This creates false negatives in training data: samples labelled as negative simply because conversion has not yet been observed. This is called the Delayed Feedback problem — the core challenge in CVR prediction.
- Existing graph-based recommenders (e.g. DGSR) rely solely on ID embeddings, carrying no semantic information — the model has no knowledge of what an item is, its category, or its attributes.
- This dissertation addresses both problems simultaneously.

2. What Is the Proposed Solution?

A three-layer architecture that integrates LLM representations, dynamic graph modelling, and delayed feedback correction:

Layer 1 — LLM Encoder

Encode item and user textual descriptions using a pre-trained language model (BERT or LLaMA). The resulting semantic embeddings replace the meaningless ID vectors used in conventional models.

Layer 2 — Dynamic Graph (DGSR-based)

Merge all user interaction sequences into a single dynamic graph with timestamps and order information. Graph neural network aggregation captures cross-user collaborative signals and long-/short-term preferences.

Layer 3 — Delayed Feedback Correction

A modified training objective (drawing from DEFER / FSIW) that accounts for false-negative labels, reducing their disruptive effect on parameter updates during training.

3. Research Questions

- RQ1** Can LLM-derived semantic embeddings replace ID-based embeddings in dynamic graph networks for CVR prediction, and what are the performance gains?
- RQ2** Do LLM-based semantic representations help the model better handle false negatives caused by delayed feedback?
- RQ3** What is the computational trade-off between the performance gains and the overhead introduced by LLM representations?

4. Dissertation Chapter Structure

Introduction

Problem background, research motivation, and the three Research Questions.

Literature Review

Delayed feedback methods (DFM, ES-DFM, FSIW, DEFER) · Graph networks (LightGCN, TGN, DGSR) · LLM-based recommendation (BERT4Rec, UniRec)

Methodology

Full architecture design: LLM Encoder + Dynamic Graph Module + Delayed Feedback Correction Loss.

Experiments

Dataset (Ali-CCP / Criteo), baseline comparisons, ablation study (each module isolated).

Results & Analysis

Quantitative answers to RQ1–3; per-module contribution analysis.

Conclusion

Summary of contributions, limitations, and directions for future work.

5. In One Sentence

This dissertation injects the semantic understanding of large language models into a dynamic graph recommender network, while correcting label noise from delayed conversions, to improve CVR prediction